

Bioinspired Stabilizing System: Dynamic Modeling and Control of a Gimbal System

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Abstract—This study investigates the design, dynamic modeling, and control of a bioinspired stabilizing system, with the aim of enhancing its accuracy and stability. The system is developed using 3D modeling software and consists of three FB5311M digital servo motors for 3-axis control (with feedback). A stable base is designed to house the MPU6050 sensor, Arduino microcontroller, and a battery to power the system. By developing a mathematical model, the relationship between the servo motor movements and the system's orientation is established. Experimental validation is conducted to assess the system's performance in maintaining precise orientation control. The results demonstrate the effectiveness of the developed model in achieving accurate and reliable operation of the bioinspired stabilizing system.

Keywords: Bioinspired Stabilizing System, Dynamic Modeling, Arduino UNO, LabVIEW

I. INTRODUCTION

The bioinspired stabilizing system under investigation serves as a critical tool in various applications, ranging from camera stabilization to educational robotics. Accurate orientation control and stability are essential for ensuring the efficient operation of such systems. In this study, we focus on the design, dynamic modeling, and control of a 3D-printed bioinspired stabilizing system utilizing three FB5311M digital servo motors for 3-axis control.

One of the primary challenges in developing a bioinspired stabilizing system is achieving precision and responsiveness in maintaining orientation. Variations in external conditions, such as sudden movements or vibrations, can affect system performance. To address these challenges, the proposed system incorporates an MPU6050 sensor for real-time orientation measurement and uses an Arduino microcontroller to process the sensor data and control the servo motors.

The bioinspired stabilizing system is designed and 3D-printed to ensure compatibility with its components, including the MPU6050 sensor, Arduino board, and power supply (Li-ion batteries). A mathematical model is developed to describe the dynamics of the system, and the Arduino-based control strategy is implemented to regulate servo movements based on the yaw, pitch, and roll angles.

Experimental validation of the proposed system is conducted to evaluate its performance in maintaining stable orientation. The results demonstrate the effectiveness of the developed model and control strategy in achieving precise and reliable operation. Additionally, the project offers an accessible and educational platform for learning about dynamic modeling, control systems, and 3D printing technologies.

II. SYSTEM ANALYSIS

The bioinspired stabilizing system consists of the following components:

- **FB5311M Digital Servo Motors:** Three digital servo motors provide 3-axis control (yaw, pitch, and roll) for precise orientation adjustments.
- **MPU6050 Sensor:** A 6-axis IMU sensor used to measure angular velocity and acceleration, providing real-time orientation data.
- **Arduino Uno Microcontroller:** Interfaces with the MPU6050 sensor and servo motors, processing orientation data and controlling motor movements.
- **CAD Model:** A custom-designed CAD model ensures proper alignment and integration of all components, including the servo motors, MPU6050 sensor, Arduino Uno, and battery holder.

III. FIGURES



Fig. 1. FB5311M Digital Servo Motor

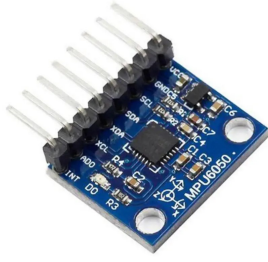


Fig. 2. MPU6050 IMU Sensor



Fig. 3. CAD Model of the Bioinspired Stabilizing System



Fig. 4. Arduino UNO

IV. MATHEMATICAL MODEL OF THE SYSTEM

The mathematical modeling involves linearizing the system for simplicity. Assume the gimbal's initial position is upright ($\theta = 0$ at this position).

A. Energy-Based Modeling

Using the energy method, the Lagrangian L is defined as:

$$L = K - P$$

where: - K is the kinetic energy. - P is the potential energy.

For the Bioinspired Stabilizing System: Dynamic Modeling and Control of a Gimbal System, the kinetic energy K is:

$$K = \frac{1}{2}j_p\dot{\theta}_p^2 + \frac{1}{2}j_y\dot{\theta}_y^2 + \frac{1}{2}j_r\dot{\theta}_r^2$$

and the potential energy P is:

$$P = mgl \cos \theta_p \approx mgl\theta_p \quad (\text{for small angles})$$

Thus, the Lagrangian becomes:

$$L = \frac{1}{2}j_p\dot{\theta}_p^2 + \frac{1}{2}j_y\dot{\theta}_y^2 + \frac{1}{2}j_r\dot{\theta}_r^2 - mgl\theta_p$$

B. Euler-Lagrange Equations

The Euler-Lagrange equations are:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_i} \right) - \frac{\partial L}{\partial \theta_i} = \tau_i$$

where τ_i is the torque applied to the i -th axis ($i = p, y, r$).
For the pitch axis (θ_p):

$$j_p\ddot{\theta}_p + mgl\theta_p = \tau_p$$

For the yaw axis (θ_y):

$$j_y\ddot{\theta}_y = \tau_y$$

For the roll axis (θ_r):

$$j_r\ddot{\theta}_r = \tau_r$$

C. Linearization

The equations are already linearized using small-angle approximations ($\sin \theta \approx \theta$, $\cos \theta \approx 1$).

D. State-Space Representation

The state-space representation of the Bioinspired Stabilizing System: Dynamic Modeling and Control of a Gimbal System is derived as follows:

$$\dot{X} = AX + BU$$

$$Y = CX + DU$$

where: - X is the state vector. - U is the input vector. - Y is the output vector.

The state vector X is defined as:

$$X = \begin{bmatrix} \theta_p \\ \dot{\theta}_p \\ \theta_y \\ \dot{\theta}_y \\ \theta_r \\ \dot{\theta}_r \end{bmatrix}$$

The input vector U is defined as:

$$U = \begin{bmatrix} \tau_p \\ \tau_y \\ \tau_r \end{bmatrix}$$

The state-space matrices are:

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{mgl}{j_p} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{j_p} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{j_y} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{j_r} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

E. Combined State-Space Equations

The state-space equations are:

$$\dot{X} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -\frac{mgl}{j_p} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} X + \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{j_p} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \frac{1}{j_y} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \frac{1}{j_r} \end{bmatrix} U$$

$$Y = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} X + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} U$$

F. Interpretation

- The matrix A describes the system dynamics, including the restoring torque due to gravity for the pitch axis. - The matrix B maps the input torques to the system states. - The matrix C selects the angular positions as outputs. - The matrix D is zero because there is no direct feedthrough from inputs to outputs.

V. DESIGN RATIONALE

The design of the Bioinspired Stabilizing System focuses on achieving precise and stable control while ensuring safety and robustness. The system utilizes a feedback control loop to continuously monitor and adjust the orientation of the gimbal.

A. Key Elements in the Design

- **Feedback Loop:** Maintains the system's orientation at the desired setpoint by continuously comparing the actual orientation (measured by sensors) with the desired orientation.
- **Decoupled Control:** Each axis (pitch, yaw, and roll) is controlled independently, simplifying the control design and implementation.
- **Safety Mechanisms:** Includes limits on angular velocities and positions to prevent mechanical damage, as well as fault detection for sensor or motor failures.

B. Control System Architecture

The control system architecture consists of the following components:

- **Sensors:** Measure the angular position and velocity of each axis (IMUs).
- **Actuators:** Apply torques to each axis using motors (servo motors).
- **Controller:** Computes the optimal control inputs using the LQR algorithm and sends commands to the actuators.
- **Setpoint:** The desired orientation is provided by the user or an external system (e.g., a camera stabilization system).

This design ensures precise and stable control of the Bioinspired Stabilizing System: Dynamic Modeling and Control of a Gimbal System, making it suitable for applications such as camera stabilization, drone navigation, and robotic systems.

VI. SIMULATION INSIGHTS

The simulation in LabVIEW ensures proper functionality before practical implementation. Key components in the LabVIEW block diagram include:

A. Serial Communication and Servo Initialization

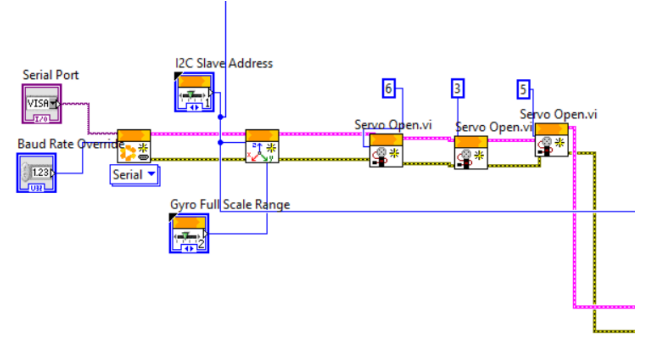


Fig. 5. Serial Communication and Servo Initialization

The blocks configure serial communication, set up the I2C slave address, and initialize servos for motor control.

B. IMU Calibration and Control Signal Processing

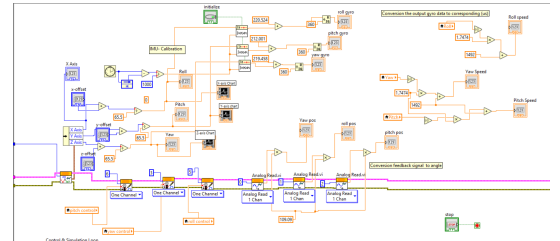


Fig. 6. IMU Calibration and Control Signal Processing

This section visualizes the calibration of the IMU (Inertial Measurement Unit), including offset adjustments for the X, Y, and Z axes. It processes gyroscopic data to determine roll,

pitch, and yaw angles and their respective speeds. The outputs are integrated into control signals for the system's stabilization and feedback mechanisms.

C. Roll Control PID Implementation

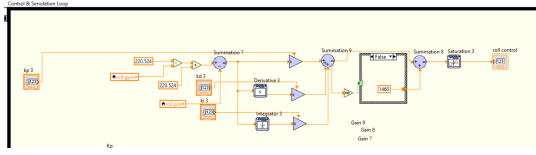


Fig. 7. Roll Control PID Implementation

This section demonstrates the PID control logic for stabilizing the roll axis. It calculates the error between the desired and actual roll positions and processes it through proportional, integral, and derivative components. The resulting control signal is adjusted for system dynamics using gain values and a saturation block, ensuring smooth and stable roll control.

D. Yaw Control PID Implementation

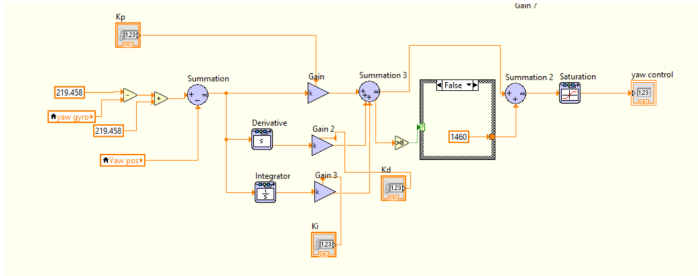


Fig. 8. Yaw Control PID Implementation

This section visualizes the PID control structure for managing the yaw axis. It calculates the yaw error based on the difference between the desired and actual yaw positions. The error is processed through proportional, integral, and derivative terms, which are scaled by respective gain values. The combined control signal is regulated using a saturation block to ensure stability and precision in yaw axis control.

E. Pitch Control PID Implementation

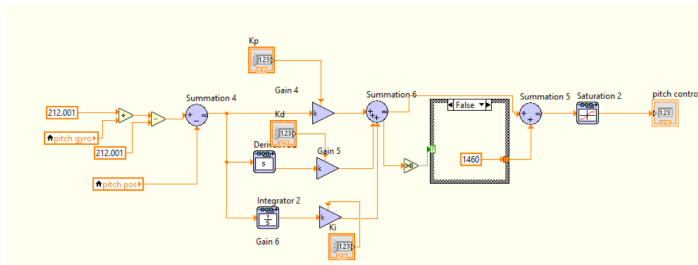


Fig. 9. Pitch Control PID Implementation

This section visualizes the PID control structure for managing the pitch axis. It calculates the pitch error based on the

difference between the desired and actual pitch positions. The error is processed through proportional, integral, and derivative terms, each scaled by respective gain values (K_p , K_i , K_d). The combined control signal is regulated using a saturation block to ensure stability and precision in pitch axis control.

F. State-Space Controller Design with LQR Optimization

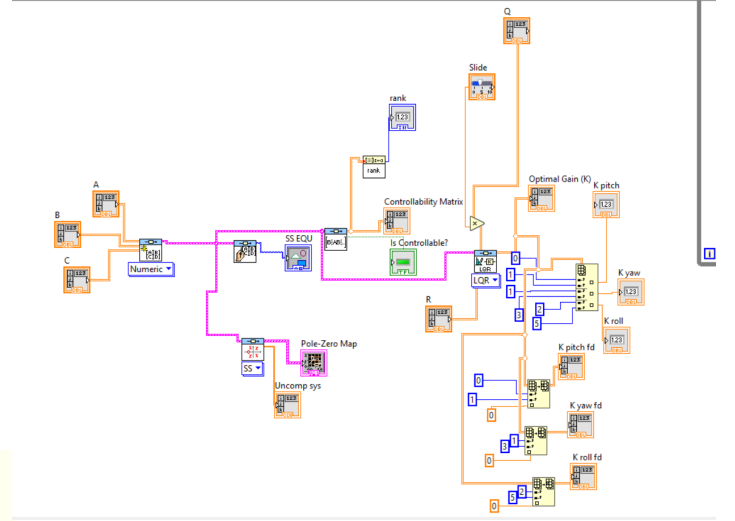


Fig. 10. State-Space Controller Design with LQR Optimization

Diagram implements a state-space representation and Linear Quadratic Regulator (LQR) optimization to design a control system. The inputs A , B , and C define the state-space matrices, while controllability is verified using the controllability matrix and rank calculation. The system's poles and zeros are analyzed, and the LQR block computes optimal feedback gain (K) based on weighting matrices Q and R , which are then divided into components for yaw, pitch, and roll control gains.

VII. RESULTS

Initially, the control system for the Bioinspired Stabilizing System was designed using a Linear Quadratic Regulator (LQR). The LQR was chosen for its optimal control strategy, which minimizes a cost function balancing performance and control effort. The initial implementation of LQR used the full state vector, including angular positions and velocities, to compute the control inputs. Despite its theoretical advantages, practical challenges arose during testing.

It was observed that the placement of the MPU6050 sensor at the final position on the gimbal introduced additional dynamics and disturbances that were not effectively handled by the LQR. This sensor placement required a more responsive and adaptable control strategy to maintain precise orientation under varying conditions.

To address these challenges, the controller was redesigned using a Proportional-Integral-Derivative (PID) control strategy. The PID controller was tuned to optimize the system's performance by adjusting its proportional (K_p), integral (K_i), and derivative (K_d) gains. Fine-tuning was performed iteratively

based on experimental feedback to achieve the best possible stability and responsiveness.

The following steps were undertaken during the transition:

- **Controller Redesign:** The LQR was replaced with a PID controller to account for the real-time dynamics and disturbances caused by the placement of the IMU.
- **Fine-Tuning of Gains:** The PID gains (K_p , K_i , K_d) were adjusted iteratively through trial and error to minimize overshoot, settle time, and steady-state error.
- **Validation:** The redesigned system was tested extensively, demonstrating significant improvements in maintaining orientation and stability under varying conditions.

The PID controller's ability to adapt to the system's dynamics and its straightforward implementation made it a practical choice for real-world applications. Experimental results confirmed that the PID controller provided reliable and precise stabilization, effectively addressing the limitations encountered with the LQR.

A. Key Findings

- **Performance Comparison:** The LQR struggled to handle the additional dynamics introduced by the IMU's placement. In contrast, the PID controller proved more effective in achieving stable and precise orientation control.
- **Tuning Sensitivity:** The system's performance was highly sensitive to the PID parameters, underscoring the importance of fine-tuning for optimal results.
- **Robustness:** The PID controller demonstrated robustness in handling disturbances and varying conditions, outperforming the LQR in practical scenarios.

This transition highlights the importance of selecting an appropriate control strategy based on the system's physical configuration and real-world requirements. Future work could focus on automating the PID tuning process or exploring adaptive PID controllers to further enhance performance.

VIII. APPLICATIONS AND FUTURE IMPROVEMENTS

A. Applications

The Bioinspired Stabilizing System developed in this project has wide-ranging applications across various fields. Key applications include:

- **Camera Stabilization:** The system enhances stability in cameras and drones, compensating for unwanted movements and vibrations. This ensures smooth, high-quality video recording in dynamic environments.
- **Robotics:** It can be employed in robotic arms or platforms to improve precision and control during tasks like assembly, inspection, and manipulation in both educational and industrial contexts.
- **Virtual Reality (VR) and Augmented Reality (AR):** By stabilizing the user's view, the system reduces motion sickness and enhances the immersive experience in VR/AR applications.

- **Medical Devices:** The system ensures stability and precision in medical equipment, such as surgical robots and handheld diagnostic tools, enabling safe and accurate operation during critical procedures.
- **Aerospace:** It stabilizes sensors, cameras, or communication devices on satellites, drones, and aircraft, ensuring reliable data collection and transmission in challenging conditions.

B. Future Improvements

The current system demonstrates effectiveness and robustness, but there are several avenues for further enhancement:

- **Integration of Advanced Sensors:** Incorporating sensors such as gyroscopes, accelerometers, and magnetometers can improve real-time detection and compensation for complex movements.
- **Machine Learning Algorithms:** Adding machine learning capabilities can make the system more adaptable and predictive, enabling it to learn from past disturbances and respond proactively to new ones.
- **Wireless Communication:** Utilizing wireless technologies like Wi-Fi or Bluetooth allows remote monitoring and control, facilitating integration into IoT (Internet of Things) platforms.
- **Energy Efficiency:** Optimizing power consumption with low-power microcontrollers and energy-efficient components can make the system more suitable for portable or solar-powered applications.
- **Modular Design:** A modular and scalable design enables adaptation to different sizes, configurations, and applications, broadening the system's usability.
- **Enhanced Control Algorithms:** Implementing advanced control strategies such as adaptive control can further improve precision, stability, and responsiveness.

IX. CONCLUSION

This study successfully presents the dynamic modeling and control of a Bioinspired Stabilizing System using servo motors and an MPU6050 sensor. By developing a comprehensive mathematical model and implementing a robust feedback control loop, the system achieves accurate and stable orientation control.

Experimental validation using Arduino UNO and LabVIEW confirms the system's effectiveness and the reliability of the control strategy. This implementation demonstrates potential applications in diverse fields, ensuring precision and efficiency in stabilization tasks. Future work will focus on integrating advanced sensors, improving energy efficiency, and leveraging machine learning-based control strategies to further enhance system performance and adaptability.